

Benchmark Metrics

Nine models for Austrian imbalance-price forecasting, measured under identical leak-free conditions.

Market APG control area (AT) · Data 140,249 quarter-hours, 2022–2025 · Test 2025, 4 rolling folds, 34,521 origins × 96 horizons · Seed 42 · Date 31 Aug 2026

1. Conditions

Every model was trained on identical rolling quarterly folds with identical scaling, identical known-future covariates, identical day-ahead outage dropout, and the day-ahead **gate-closure fix enabled** — so no model sees an auction price that had not yet cleared at its forecast origin. The design is a 2 × 3 grid (market-rule blocks against a plain mixer, across Dense, LSTM and GRU encoders) plus four references. Reading down a column isolates the encoder; reading across a row isolates the market rules.

All accuracy figures are **pooled** on concatenated per-fold predictions, each already inverted to EUR/MWh inside its own fold. This matters: RMSE is a root of a mean of squares and does not average across folds — the mean of four fold-RMSEs understates the pooled value by roughly 17%.

2. Results

MODEL	RULES	AQL	MAE	RMSE	SMAPE	MASE	AQCR	PARAMS	VS CLIM AQL	VS CLIM MAE	TRAIN S	VAL S	TEST S	TOTAL S
MRINN	yes	25.82	66.22	389.97	63.18	0.68	0.00	6,135	−11.43%	−11.94%	218.8	20.1	0.66	244.4
MR-LSTM	yes	25.67	65.28	389.37	62.74	0.67	0.00	5,895	−11.94%	−13.19%	409.5	85.3	0.90	498.9
MR-GRU	yes	25.57	65.01	389.37	62.24	0.67	0.00	5,055	−12.27%	−13.56%	625.0	109.8	0.88	738.9
MLP	no	25.87	66.17	389.63	63.53	0.68	0.00	7,533	−11.25%	−12.01%	119.6	29.4	0.40	154.3
LSTM	no	25.75	66.20	389.61	63.90	0.68	0.00	7,261	−11.68%	−11.97%	320.4	50.5	0.68	374.8
GRU	no	25.86	66.79	390.04	64.30	0.69	0.00	6,309	−11.29%	−11.18%	424.7	58.9	0.66	487.5
AttnBiLSTM	no	25.83	66.52	389.63	63.93	0.69	0.00	15,302	−11.39%	−11.55%	749.0	100.8	1.04	854.1
XGBoost	no	26.35	67.44	389.94	64.63	0.70	2.75	—	−9.62%	−10.32%	2,807.0	—	0.49	2,810.6
LQR	no	28.02	74.96	391.78	72.13	0.78	18.07	538,080	−3.87%	−0.33%	358.6	134.2	0.19	497.8

Reference baselines on the same rows: **climatology** AQL 29.150, MAE 75.20, RMSE 393.8 · **seasonal naive** AQL 39.870, MAE 96.03, RMSE 547.0. Times are seconds per fold; multiply by four for the whole study.

3. What each column means

METRIC	UNITS / RANGE	DEFINITION AND WHY IT IS HERE
AQL	EUR/MWh, lower better	Average quantile loss — mean pinball loss over the five quantiles (0.1, 0.25, 0.5, 0.75, 0.9). This is the headline metric because the model is probabilistic: it scores the whole predicted distribution, not just the central estimate.
MAE	EUR/MWh, lower better	Mean absolute error of the median (q50) forecast. Interpretable in market units and robust to the extreme tail.
RMSE	EUR/MWh, lower better	Root mean squared error of q50. Squaring makes it dominated by spikes — the imbalance price reaches ±15,000 EUR/MWh — which is why all models sit near 390 regardless of skill.
sMAPE	%, 0–200, lower better	Symmetric mean absolute percentage error. Included on request; see the caveat below.
MASE	ratio, lower better	Mean absolute scaled error, scaled by the seasonal-naive change at season = 96. Scale-free and, unlike sMAPE, valid with negative prices. Below 1 means better than seasonal naive — every model qualifies.

METRIC	UNITS / RANGE	DEFINITION AND WHY IT IS HERE
AQCR	%, lower better	Average quantile crossing rate — how often predicted quantiles cross (e.g. q75 below q50), which is logically impossible and signals a broken forecast distribution.
Params	count	Trainable weights. A proxy for model capacity and overfitting risk.
vs clim	%, negative better	Improvement over climatology — predicting the training-set quantiles for every interval. This is the honest reference: beating a seasonal naive on this series is easy, beating climatology is not.
Train / Val / Test	seconds per fold	Computational cost split into fitting, per-epoch validation, and inference over the test set.

Caveat on sMAPE. It is computable here — the denominator $|y| + |\hat{y}|$ never vanishes — but it **saturates**. 6.8% of target cells are negative, and **8.8% sit at the 200% ceiling**, where the metric returns the same value whether the miss is 100 or 10,000 EUR/MWh. Those cells are 9.2% of the data but 29.6% of the metric, and their mean absolute error is 174 EUR/MWh. sMAPE therefore cannot rank tail behaviour, which is what this market turns on. It ranks the models in the same order as AQL, so it is not misleading as an ordering — but its magnitude should not be interpreted. MASE is reported beside it for that reason.

Note on timing. Train and validation are the measured per-epoch cost multiplied by the recorded epoch count. Per-epoch cost is taken as a *slope* — differencing a 6-epoch fit against a 2-epoch fit — so the fixed graph-tracing and setup overhead cancels rather than being multiplied by the epoch count. Test time and data preparation are measured directly. Reconstructed totals land 8–12% below the originally recorded per-fold wall time, which is expected: that figure also covered inverse-transforming, per-horizon scoring and writing prediction archives. **XGBoost has no epochs**, so its training figure is a direct measurement of the full 96-booster fit and validation does not apply; its `Params` is blank because the stored value was 2,993 boosting rounds, not weights, and printing that beside a neural weight count would invite a false comparison.

4. The same models at the three reporting horizons

The pooled table averages over all 96 horizons and hides a genuine crossover.

MODEL	AQL 15 MIN	AQL 1 HOUR	AQL 1 DAY	MAE 15 MIN	MAE 1 HOUR	MAE 1 DAY	SMAPE 15 MIN	SMAPE 1 HOUR	SMAPE 1 DAY
MRINN	25.14	25.38	26.11	64.17	64.78	67.19	61.75	61.87	63.80
MR-LSTM	25.18	25.36	26.08	63.15	63.80	66.90	61.36	61.64	63.43
MR-GRU	24.92	25.18	25.92	62.20	63.27	66.32	60.27	60.62	62.72
MLP	25.28	25.37	26.26	64.21	64.34	67.42	62.38	61.98	64.60
LSTM	24.85	25.20	25.96	63.10	64.24	66.99	61.25	61.66	64.35
GRU	24.57	25.19	26.01	61.83	64.11	67.05	60.92	61.79	64.07
AttnBiLSTM	25.17	25.36	26.34	64.19	64.66	68.10	62.86	62.36	65.32
XGBoost	22.97	25.48	26.88	56.35	63.59	68.53	55.34	61.04	65.42
LQR	25.91	27.33	28.59	67.69	73.59	77.06	65.72	70.02	72.92

5. Statistical significance

Diebold–Mariano, computed two ways because adjacent origins share 95 of their 96 targets: Newey–West with lag 96 so the overlap sits inside the bandwidth, and non-overlapping daily origins (one per day, 361 points) where the classical assumption holds outright. Negative favours the first-named model.

COMPARISON	ISOLATES	Δ AQL	P (NW)	P (DAILY)	VERDICT
MR-GRU – MRINN	encoder, rules held	−0.245	0.0002	0.0017	GRU encoder wins
MR-GRU – MR-LSTM	encoder, rules held	−0.096	0.0002	0.0014	GRU encoder wins
MR-GRU – GRU	rules, GRU encoder	−0.286	0.0000	0.0002	rules help
MR-LSTM – LSTM	rules, LSTM encoder	−0.075	0.153	0.961	not significant
MRINN – MLP	rules, Dense encoder	−0.052	0.417	0.255	not significant

6. How to read the table

MR-GRU is the best model, and it is also the smallest

It wins every accuracy column while using **5,055 parameters** — the fewest of any neural model, a third of AttnBiLSTM's and a hundredth of LQR's — and completes the four-fold study in 49 minutes against AttnBiLSTM's 57 and XGBoost's 187. That is a clean accuracy-per-parameter and accuracy-per-second argument. Inference is under one second per fold for all nine models, so the rule blocks impose no deployment cost, only training cost.

The encoder swap is real but small

MR-GRU beats MRINN by 0.245 AQL, significant on both tests — a genuine effect, but a **0.95% improvement**. Changing the encoder alone is a thin contribution.

The market rules are worth about 0.53% — and only at longer horizons

Averaged across the three encoders the rule blocks buy roughly half a percent, and only the GRU pairing reaches significance. The per-horizon table shows why the pooled figure misleads: **at 15 minutes the rules actively hurt** both recurrent models (plain GRU 24.57 against MR-GRU 24.92; plain LSTM 24.85 against MR-LSTM 25.18), and the sign only flips by day-ahead. The defensible claim is therefore narrower and more interesting than "market rules help": rule structure costs accuracy where persistence is strong, and earns its place as the horizon lengthens.

XGBoost is the mirror image

It is the strongest model at 15 minutes by a wide margin (AQL 22.97, 8% clear of MR-GRU) and second-worst at a day (26.88). Its advantage is entirely short-horizon, which is the same effect seen from the other side.

Only two models can produce an invalid distribution

AQCR is exactly zero for seven of nine. That is earned by the hierarchical quantile head, which makes crossing structurally impossible, not by the encoder. XGBoost (2.75%) and LQR (18.07%) lack that head, and LQR's figure means nearly one prediction in five is internally inconsistent.

The honest headline. The best model improves on climatology by **12.3% AQL and 13.6% MAE**. Against the seasonal naive the same model looks far better, but that baseline is weak on this series and should not be used for the headline. All nine models sit within 3% of one another, which is itself a finding: the residual error is dominated by the unpredictability of the imbalance direction, not by model choice.

Reproducible from `benchmark_models.py`, `results_table.py`, `timing_benchmark.py`, `dm_test.py` and `baselines.py`. Data: APG control area, 140,249 quarter-hours, 2022–2025. Test year 2025, four rolling folds retrained quarterly, seed 42. Gate closure enabled throughout.